Fiscal Year:	FY 2018	**Task Last Updated:**	FY 11/18/2018
PI Name:	Turner, Russell T Ph.D.		
Project Title:	Spaceflight-Induced Changes in Non-Shivering Thermogenesis and Effects on Bone in Mice		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	97331-8558	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	07/01/2015	**End Date:**	06/30/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 6/30/2018 per F. Hernandez/ARC (Ed., 6/30/17) NOTE: Extended to 6/30/2017 per F. Hernandez/ARC (Ed., 5/20/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Branscum, Adam Ph.D. (Oregon State University) Iwaniec, Urszula Ph.D. (Oregon State University) Wong, Carmen Ph.D. (Oregon State University)		
Grant/Contract No.:	NNX15AL15G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The following tissues are requested from mice flown on the International Space Station (ISS) and their respective ground controls: (1) interscapular brown adipose tissue (BAT), (2) hypothalamus, (3) intraperitoneal white adipose tissue (WAT), and (4) left and right femora. The objective of this project is to test the hypothesis that spaceflight will compromise the ability of mice to employ huddling and postural strategies to slow heat loss. To maintain body heat, flight animals will become more dependent on a mechanism, non-shivering thermogenesis, which is associated with bone loss. Methods: mRNA levels for uncoupling protein-1 (UCP-1) will be measured in BAT as a sensitive index of non-shivering thermogenesis. mRNA levels for leptin receptor and NPY (regulators of non-shivering thermogenesis) will be measured in hypothalamus and mRNA levels for enzymes involved in fat metabolism will be measured in BAT and WAT. We anticipate that fat metabolism will be increased during spaceflight in response to increased sympathetic outflow (a negative regulator of bone formation). We will measure bone microarchitecture and cells in the right femur and mRNA levels for bone genes important to mesenchymal and hematopoietic cell differentiation in the left femur. We anticipate that spaceflight will result in lower cancellous bone mass, decreased osteoblast number, and decreased expression of genes important for osteoblast differentiation. Furthermore, spaceflight will result in increases in bone marrow adipocytes and osteoclasts, and increases in expression of genes important to adipocyte and osteoclast differentiation. Significance: The proposed research will address the mechanism(s) by which the spaceflight environment impacts energy balance and bone turnover. At the completion of this study we expect to show that spaceflight results in increased non-shivering thermogenesis and bone loss. The primary positive impact of our anticipated findings is evidence for a novel mechanism for spaceflight-induced bone loss in mice. Specific Aim 1: Measure uncoupling protein-1 (UCP-1) gene expression in interscapular brown adipose tissue (BAT). UCP-1 is a sensitive index of non-shivering thermogenesis. If our hypothesis is correct, mRNA levels of UCP-1 will be higher in the BAT from flight mice compared to ground controls. We will also measure mRNA levels for leptin receptor and neuropeptide-Y (NPY) in the hypothalamus. Leptin signaling (increases) and NPY signaling (decreases) in the hypothalamus have opposing effects on UCP-1 expression in BAT. Specific Aim 2: Measure mRNA levels for genes associated with fat metabolism in intraperitoneal WAT. If our hypothesis is correct, mRNA levels for genes associated with fat metabolism will be increased in flight mice compared to ground controls, reflecting the metabolic response to increased sympathetic outflow. Specific Aim 3: Measure bone mass, density, microarchitecture, and cells (osteoblasts, osteoclasts, and adipocytes) in the right femur and mRNA levels for bone matrix proteins and genes associated with mesenchymal (osteoblast, osteoclast, adipocyte) and hematopoietic cell differentiation and function in the left femur. If our hypothesis is correct, flight animals will have lower cancellous bone mass and the bone loss will be associated with decreased osteoblast number and decreased expression of genes associated with osteoblast differentiation. Furthermore, spaceflight will result in increases in bone marrow adipocytes and osteoclasts, and increases in expression of genes associated with adipocyte and osteoclast differentiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A minute imbalance between energy intake and expenditure (<1%) can lead to massive changes in body weight (anorexia and obesity). Non shivering thermogenesis is the principle mechanism balancing energy intake and expenditure. We believe that this research will improve understanding of the regulation of non shivering thermogenesis and factors that lead to its dysregulation.
Task Progress:	Summary of Accomplishments Progress on Specific Aim 1 1) We performed ground-based proof of principle studies which successfully demonstrated the ability to isolate high quality mRNA from brown fat obtained from partially thawed carcasses from mice that had been frozen on dry ice and stored at -80ºC, avoiding the requirement for necropsy of brown adipose tissue on ISS. 2) We requested and received brown adipose tissue from RR1 B6 mice flown aboard ISS and age- and housing-matched ground controls. mRNA was successfully isolated from flight and ground controls animals. To date, we have performed targeted gene array analysis on these samples to identify differentially expressed genes related to adipogenesis, fatty acid metabolism, and AMPK signaling. 3) We received brown adipose tissue from RR3 BALB/c mice flown aboard ISS and ground controls. mRNA was successfully isolated from flight and ground control animals. To date, we have performed targeted gene array analysis on these samples to identify differentially expressed genes related to adipogenesis. Progress on Specific Aim 2. We performed ground-based proof of principle studies successfully demonstrating our ability to isolate high quality mRNA from white adipose tissue obtained from partially thawed carcasses from mice that had been frozen on dry ice and stored at -80ºC, avoiding the requirement for necropsy of brown adipose tissue on ISS. 1) We received white adipose tissue from RR1 B6 mice flown aboard ISS and their ground controls. Unfortunately, the RNA was degraded. This likely occurred during the transfer of white adipose tissue excised from partially thawed mice into RNAlater. We performed studies which confirmed that transfer of fresh tissue into RNAlater is preferable to freezing, followed by transfer into RNAlater. 2) We received frozen white adipose tissue from RR3 BALB/c mice flown aboard ISS and ground controls. We successfully isolated high quality mRNA from these samples. To date, we have performed targeted gene array analysis on these samples to identify differentially expressed genes related to adipogenesis. 3) We have received archived frozen white adipose tissue from RR1. To date, we have performed targeted gene array analysis on these samples to identify differentially expressed genes related to adipogenesis Progress on Specific Aim 3 Hypothalamus and bone specimens from flight experiments were unavailable. However, we have performed preliminary analysis using archived tissues from hindlimb unloading studies supported by NNX15AL15G and department funds. This analysis will complement the spaceflight data described above.

	Significance The data to date provide strong evidence that non-shivering thermogenesis is increased in B6 mice by spaceflight. More than 30 genes related to adipogenesis, fatty acid metabolism, or AMPK metabolism were differentially expressed. Mice employ a variety of strategies to decrease reliance on non-shivering thermogenesis to successfully adapt to room temperature housing. These include nest building, huddling, and postural changes. Our data to date supports the hypothesis that the spaceflight environment compromises the ability of mice to employ these strategies to slow heat loss, and as a consequence flight animals become more dependent upon non-shivering thermogenesis to maintain body heat. A manuscript is under review and several more will be submitted later in 2018.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Turner RT, Branscum AJ, Wong CP, Iwaniec UT, Morey-Holton E. "Studies in microgravity, simulated microgravity and gravity do not support a gravitostat." J Endocrinol. 2020 Dec;247(3):273-82. https://doi.org/10.1530/JOE-20-0393 ; PMID: 33108334 , Dec-2020
Articles in Peer-reviewed Journals	Gamboa A, Branscum AJ, Olson DA, Sattgast LH, Iwaniec UT, Turner RT. "Effects of spaceflight on cancellous and cortical bone in proximal femur in growing rats." Bone Rep. 2021 Feb 14;14:100755. https://doi.org/10.1016/j.bonr.2021.100755 ; PMID: 33665238; PMCID: PMC7907224 , Feb-2021
Articles in Peer-reviewed Journals	Wong CP, Iwaniec UT, Turner RT. "Evidence for increased thermogenesis in female C57BL/6J mice housed aboard the International Space Station." npj Microgravity. 2021 Jun 18;7(1):23. https://doi.org/10.1038/s41526-021-00150-y ; PMID: 34145277 , Jun-2021
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Articles in Peer-reviewed Journals	Martin SA, Riordan RT, Wang R, Yu Z, Aguirre-Burk AM, Wong CP, Olson DA, Branscum AJ, Turner RT, Iwaniec UT, Perez VI. "Rapamycin impairs bone accrual in young adult mice independent of Nrf2." Exp Gerontol. 2021 Oct 15;154:111516. Epub 2021 Aug 10. https://doi.org/10.1016/j.exger.2021.111516 ; PMID: 34389472 , Oct-2021